What This Study Can and Cannot Tell You

Applying the trajectories data to a PhD-or-build decision

23 August 2026 · revised after your corrections

Read this before any number below

You collected this data, so you know the caveat, but it governs everything here: **the median is a duration, not a probability.** It says how long the road ran for people who reached the end. It says nothing about how many set out.

This study therefore **cannot** tell you whether a PhD or a business is more likely to work. It never sampled anyone who tried either and failed. Both of your options appear here only through their winners. I will not write a sentence of the form "the data says path A beats path B," because no such sentence is supportable.

What it *can* do is describe the timing and shape of careers that worked, and on your specific question it turns out to say something quite sharp.

1 Three corrections to my first draft

You caught three things, and two of them changed the conclusion.

You are 24, not 22. Born May 2002. That shortens the runway to the median but does not change its shape — see below.

The visa argument is withdrawn. My first draft leaned on immigration asymmetry as the tiebreaker. You closed that question on 1 August 2026: no revenue means no exposure, and a Korean entity can take revenue lawfully if it comes to that. Both points are sound, the ruling is recorded in `business/notes`, and I should not have re-opened it. The argument is removed rather than softened, because it was also weaker than I made it sound.

You do not want to be a physics researcher. That removes the entire academic-prize comparison from my first draft as irrelevant to you. It matters in a second way too: it means the PhD, if you do one, is not a career — it is a five-to-six-year input into something else. That reframes what it has to be worth, and the data has an opinion about it.

2 The runway

You are 24. The median person here who eventually crossed $10M constant did it at 33 — nine years out. A quarter of them did it after 39. You are not late, and there is no closing window. That is the first thing worth internalising, because a decision made from urgency is worse than the same decision made calmly, and nothing in this data supports the urgency.

Quantity	Median	Spread
Age at first hit	**33.0**	middle half 30–39, n=105
if the hit landed post-1995	32.0	middle half 28.8–36.0
Years from age 18 to first hit	15.0	middle half 12–21
Years from starting the venture that worked	**4.2**	middle half 3.0–7.2

2.1 The company is the short part

Put two of those numbers together. From 18 to the hit is a median of 15 years. From *starting the venture that worked* to the hit is 4.2.

> **Roughly eleven of those fifteen years happen before the winning venture begins.**

The build is short. The long part is whatever precedes it. And because this study dates a serial founder at their **first** company to cross rather than their famous one, the earlier failed attempts are counted *inside* those eleven years, not excluded from them.

So the question is not "PhD or business." It is *what the next decade is made of*, given that the company itself is only about four years of it.

3 The finding that bears directly on your choice

I built a profile table for all 235 people — highest completed degree, field, institution — and cut it against commercial hits only. Prizes are excluded entirely, so this is people who built a \$10M-constant business, which is the thing you are considering.

Highest completed degree	n	Median age at hit	Venture clock
PhD or MD	14	**43.5**	4.5 yr
Masters	31	33.8	4.0 yr
Bachelors	45	**31.0**	4.0 yr
High school or less	19	28.5–33	4.0 yr

Among people who built a \$10M business, the doctorate holders got there about **twelve years later** — and their venture clock is identical. The doctorate does not make the company slower. It moves the starting line.

Three checks, because a gap that large invites the obvious objections:

- **Is it just industry?** No. Every single PhD/MD commercial winner in this sample sits in hardware/deep tech (8) or healthcare/biotech (6). **Zero built a software or internet business.** Restricting to those slow industries alone, the gap persists: 43.5 against 32.5.
- **Is it just the longer education?** No. Measured from the end of education — which for a PhD is already five years later — doctorate holders still took **15.0 years** to the hit against **8.5** for bachelors or less. They start later *and* take longer afterwards.
- **Is the sample big enough?** No. n=14, and that is the honest limit on this finding. Treat the direction as informative and the magnitude as soft. It is also survivorship on both sides: these are the PhDs who succeeded commercially.

Set against your situation: your ideas in `~/business` — camera arbitrage, the card tracker, the recorder bundle, the storage kit, dropshipping — are consumer hardware, import and distribution plays. They sit squarely in the quadrant where the bachelors-or-less cohort hits at 31, and nowhere near the deep-tech and biotech quadrant that is the *only* place doctorate holders show up commercially in this dataset.

4 But switching fields is cheap, which cuts the other way

The counterweight, and it is a real one. I split the winners by whether they entered the domain of their eventual hit soon after finishing school, or years later.

	n	Median age at hit	Venture clock
Entered the domain within 2 years of school	59	32.5	4.0 yr
Entered it 3+ years later (career switchers)	16	34.2	4.8 yr

One in five winners switched into their field late, with a median detour of 4.5 years, and it cost them under two years at the median — a difference the intervals cannot distinguish from zero. One person switched after 28 years and still crossed.

So: doing something unrelated for several years is **not** disqualifying, and a physics PhD would not brand you as having missed your chance. This is the strongest thing the data says in favour of taking the time.

What it does not say is that the detour *helps*. The switchers did fine *despite* the gap, not because of it.

5 Where that leaves the decision

The data supports these four statements, and nothing stronger:

1. **You are not late.** Median age at hit is 33; you are 24; the middle half ran to 39. No urgency is justified.
2. **The venture is about four years. The decade before it is the variable.** Choose that decade deliberately — it is where the time goes.
3. **For the kind of business you actually want to build, the doctorate cohort is absent.** Zero of 14 PhD/MD commercial winners built software or consumer distribution. Your ideas live where the 31-year-olds are.
4. **A multi-year detour is survivable.** A fifth of the winners took one, at a cost under two years. If you want the PhD for its own sake, the data does not say it forecloses building later.

Which points at a fairly specific reading. **The PhD is defensible as a thing you want, or as a hedge you value. It is not defensible as preparation for these businesses**, because the people who built these businesses did not have one, and the ones who did took twelve years longer to get there. If you are doing research to fill a resume rather than from interest, the honest question is what the resume is for — because it is not for the camera business.

6 The test I would actually run

Not a recommendation on the decision, but on how to make it. You have until October.

The camera arbitrage and card-tracker ideas are unusually *testable*: real customers, observable margins, a short cash cycle, and no capital that you cannot walk away from. One quarter of genuine trading — actual units, actual landed cost, actual buyers — would tell you more about whether you want this life than this entire study can, and more than another year of deliberation.

That test is also cheap in exactly the dimension the data says is scarce. It consumes a few months of the eleven-year pre-venture window, and it resolves the question the data cannot: not *how long does it take*, but *is there a specific thing here worth eleven years.* Everyone in this dataset crossed with a particular product for a particular buyer. "Start a business" is not a plan, and the four-year venture clock does not apply to a category.

If the test says yes, the decision makes itself and the PhD becomes a cost. If it says no, you will have learned that before committing either way, which is worth more than either answer.

One last time, because it is the misreading this dataset invites most: nine years, or thirty-three, is how long it took the people who made it. It is not your odds, and it is not a schedule. The people who spent the same decade and got nothing are not in this file — not because they were rare, but because the study was built in a way that cannot see them.